QUEST-KG: Evidence-Aware and Uncertainty-Calibrated Graph-RAG for Trustworthy Knowledge Graph Reasoning

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Abstract

Knowledge graphs (KGs) increasingly support enterprise intelligence, retrieval-augmented reasoning, policy enforcement, and multi-hop question answering across heterogeneous and evolving data ecosystems. Existing KG reasoning systems often optimize predictive accuracy while providing limited support for provenance-aware inference, calibrated uncertainty estimation, and robust reasoning under incomplete or noisy evidence. We present **QUEST-KG**, an evidence-aware and uncertainty-calibrated Graph-RAG framework that unifies (1) a schema-aware Graph-RAG retrieval layer for grounded subgraph construction, (2) an evidential message-passing mechanism that propagates calibrated belief and uncertainty across multi-hop reasoning chains, and (3) a neuro-symbolic reasoning module for policy and access-control inference. We evaluate QUEST-KG across four datasets spanning three task families - multi-hop QA (WebQSP, ComplexWebQuestions), temporal link prediction (ICEWS18), and dynamic policy reasoning (OrgAccess) - and additionally demonstrate generalization to cross-KG entity alignment on DWY100K (DBP-YG). Under matched-LLM conditions (Qwen2.5-7B), QUEST-KG **beats every LLM-RAG baseline on all four datasets (40/40 paired-bootstrap deltas positive, 34/40 significant at p<0.05)** while running **19-500x faster** and exhibiting **3-10x better calibration (ECE)**. On DBP-YG, our text-grounded retrieval beats RREA (+0.045) and BootEA (+0.171) **without any entity-alignment training**. These results suggest that evidence-aware, uncertainty-calibrated Graph-RAG reasoning provides a scalable, trustworthy foundation for next-generation knowledge-centric AI systems.

Keywords

knowledge graphs, Graph-RAG, evidential reasoning, uncertainty calibration, neuro-symbolic reasoning, provenance-aware inference, multi-hop question answering, entity alignment, access-control reasoning, retrieval-augmented reasoning, trustworthy AI

1. Introduction

Knowledge graphs (KGs) have become a foundational infrastructure for modern information systems and knowledge-centric AI, enabling semantic search, multi-hop reasoning, enterprise intelligence, and retrieval-augmented generation (RAG) across heterogeneous data ecosystems. Recent advances in large language models (LLMs) and retrieval-augmented reasoning have accelerated interest in graph-centric retrieval pipelines that combine symbolic relational structure with neural semantic retrieval. Compared with vector-based retrieval, graph-aware retrieval better preserves relational dependencies, supports compositional reasoning, and provides interpretable evidence aggregation for knowledge-intensive tasks such as question answering, entity alignment, and policy reasoning.

Despite this progress, existing Graph-RAG systems still face challenges in trustworthy reasoning, uncertainty calibration, and provenance-aware inference. Many current approaches rely on opaque end-to-end neural pipelines that optimize predictive accuracy while providing limited support for calibrated confidence estimation, explicit evidence tracing, or robust reasoning under incomplete and dynamically evolving graph structures. These limitations become particularly problematic in enterprise and security-sensitive environments where reasoning systems must reconcile conflicting evidence, quantify uncertainty, preserve provenance, and generate interpretable reasoning traces.

We introduce **QUEST-KG**, an evidence-aware and uncertainty-calibrated Graph-RAG framework that treats graph retrieval and downstream reasoning as a unified evidential inference process with explicit uncertainty propagation and provenance-aware belief fusion. QUEST-KG integrates three tightly coupled components: a schema-aware Graph-RAG retrieval layer, a provenance-aware evidential message-passing mechanism, and a neuro-symbolic reasoning module with selective abstention.

The main contributions are:

- **Unified evidence-aware Graph-RAG framework.** A single interpretable pipeline integrating retrieval, provenance-aware reasoning, and uncertainty propagation.
- **Provenance-aware evidential reasoning.** An evidential message-passing mechanism with Dirichlet belief and uncertainty propagation modulated by semantic similarity, provenance reliability, and schema consistency.

- **Neuro-symbolic reasoning with abstention.** Path scoring with explicit symbolic constraints and a posterior-mass / entropy abstention rule for selective prediction under uncertainty.
- **Comprehensive matched-condition evaluation.** Across four benchmark datasets and three task families, QUEST-KG beats every LLM-RAG baseline (40/40 paired-bootstrap wins) under matched-LLM conditions, with 19-500x lower latency and 3-10x better calibration. Additional EA evaluation on DBP-YG beats RREA and BootEA without EA-specific training.

2. Related Work

2.1 Knowledge Graph Reasoning and Multi-Hop Inference

Early embedding-based approaches focused on relational representation learning through translational and geometric methods such as TransE, RotatE, Query2Box, and BetaE. More recent work has incorporated graph neural networks and neural message passing for compositional reasoning over large-scale knowledge graphs. Despite progress, most KG reasoning systems optimize predictive accuracy with limited support for calibrated uncertainty, provenance-aware reasoning, or interpretable evidence tracing - motivating frameworks that jointly support uncertainty-aware inference and symbolic consistency.

2.2 Retrieval-Augmented Generation and GraphRAG

Retrieval-augmented generation has improved factual grounding in large language models. Recent GraphRAG frameworks - including Microsoft GraphRAG, ToG, RoG, and ChatKBQA - combine graph traversal with neural retrieval for context-aware reasoning. However, most existing systems treat retrieval and reasoning as independent stages and lack support for uncertainty calibration and provenance-aware inference. QUEST-KG models retrieval and reasoning as a unified evidential inference process.

2.3 Uncertainty-Aware and Trustworthy Graph Reasoning

Bayesian graph neural networks, evidential deep learning, and probabilistic message passing have been explored for trustworthy inference. While these methods improve robustness on node classification and link prediction, they generally do not integrate provenance-aware retrieval, symbolic reasoning, and multi-hop evidential reasoning into a unified GraphRAG framework. QUEST-KG fills this gap.

2.4 Neuro-Symbolic Reasoning

Neuro-symbolic reasoning aims to combine the generalization of neural models with the interpretability of symbolic reasoning. Prior work has explored neuro-symbolic methods for QA and graph reasoning but largely without explicit provenance-aware evidence fusion or uncertainty calibration. QUEST-KG extends this direction by integrating neuro-symbolic constraints into an uncertainty-calibrated retrieval pipeline.

3. Methodology

3.1 Overview

QUEST-KG performs trustworthy reasoning over a knowledge graph by treating *edge admission* into a query-specific subgraph as the central decision. Given an input query *q*, QUEST-KG first retrieves a semantically grounded evidence subgraph G_q through schema-aware graph retrieval. Provenance-aware evidential message passing then propagates semantic evidence and uncertainty estimates across relational paths. Finally, a neuro-symbolic reasoning module integrates symbolic constraints with neural inference to produce a calibrated, interpretable answer with selective abstention under insufficient evidence:

$$\blacksquare = f_\theta(q, G_q, \blacksquare, \blacksquare)$$

where *P* represents provenance-aware evidential signals and *U* denotes uncertainty estimates.

3.2 Schema-Aware Graph-RAG Retrieval

Given a query *q*, QUEST-KG retrieves a subgraph that preserves relational dependencies and multi-hop contextual structure. Let $G = (V, E, R)$ denote the KG. For query embedding **q** and entity embedding $\mathbf{v}_i$, semantic relevance is:

$$s(q, v_i) = (q \cdot v_i) / (\blacksquare q \blacksquare \cdot \blacksquare v_i \blacksquare)$$

We then take the top-*K* candidates and perform bounded multi-hop graph expansion with schema consistency and provenance constraints to produce $G_q = (V_q, E_q, R_q)$. Ontology-aware relation normalization mitigates schema drift.

3.3 Provenance-Aware Evidential Message Passing

Each node v_i maintains a tuple (h_i, b_i, u_i) where h_i is the node representation, b_i the evidential belief, and u_i the uncertainty. Aggregation uses provenance-aware attention:

$$m_i = \sum_{j\in N(i)} \alpha_{ij} W_r h_j$$

where the compatibility score combines semantic, provenance, schema signals, and propagated uncertainty:

$$\phi_{ij} = \gamma_1 s^{sem}{}_{ij} + \gamma_2 s^{prov}{}_{ij} + \gamma_3 s^{sch}{}_{ij} - \gamma_4 u_j$$

Belief and uncertainty are updated through evidential fusion: $b^{(t+1)}{}_i = \lambda\, b^{(t)}{}_i + (1-\lambda)\, \blacksquare_i$, $u^{(t+1)}{}_i = 1 - b^{(t+1)}{}_i$.

3.4 Neuro-Symbolic Multi-Hop Reasoning and Abstention

Given retrieved paths Π_q, candidate chains are scored:

$$S(\pi) = \gamma_1 s_{sem}(\pi) + \gamma_2 s_{sym}(\pi) + \gamma_3 s_{prov}(\pi) - \gamma_4 u(\pi)$$

The final prediction selects the highest-confidence reasoning path; QUEST-KG abstains when posterior mass falls below $p\star$ or entropy exceeds $H\star$.

3.5 Scalability

Approximate nearest-neighbor retrieval (FAISS) enables $O(d \log|V|)$ retrieval; evidential MP operates only over $O(|E_q|d)$ retrieved edges. Bounded multi-hop retrieval, provenance-guided edge pruning, and subgraph caching keep inference efficient on commodity hardware (we report 9-78 ms / query on a single GTX 1080 Ti).

4. Experimental Evaluation

4.1 Research Questions and Setup

We evaluate QUEST-KG to answer: **RQ1** Does evidential graph reasoning improve multi-hop reasoning across heterogeneous tasks? **RQ2** Does uncertainty-aware retrieval improve calibration? **RQ3** Does the framework generalize across task families? **RQ4** Is QUEST-KG practical for deployment?

Hardware. Local symbolic runs on a single NVIDIA GTX 1080 Ti (11GB). LLM baselines on a Google Colab L4 GPU (22.5GB). **Encoder.** sentence-transformers/all-MiniLM-L6-v2 (frozen). **LLM backbone.** Qwen2.5-7B-Instruct (frozen, no fine-tuning) for all RAG baselines and the QUEST-KG-LLM hybrid.

4.2 Benchmark Datasets

Dataset	Task family	Domain	Test N
OrgAccess	Dynamic policy	Enterprise (synthetic)	750
ICEWS18	Temporal link prediction	Event KG	200
WebQSP	Multi-hop QA	Freebase	1,000
ComplexWebQuestions	Multi-hop QA	Freebase	500
DWY100K - DBP-YG	Entity alignment	DBpedia / YAGO	1,000 anchors / 100K cands

Table 1: Datasets used in evaluation. The four core datasets cover three task families (policy reasoning, temporal link prediction, multi-hop QA). We additionally evaluate cross-KG entity alignment on DBP-YG from the DWY100K family to demonstrate generalization.

4.3 Headline Results (Matched-Condition)

Table 2 reports the primary task metric for each method on each dataset. All LLM-based methods use the same frozen Qwen2.5-7B backbone. **QUEST-KG and QUEST-KG-LLM dominate every dataset.**

Method	OrgAccess (BAcc)	ICEWS18 (MRR)	WebQSP (Hits@1)	CWQ (Hits@1)	DBP-YG (Hits@1)
Vanilla-RAG	0.507	0.016	0.185	0.149	n/a
GraphRAG	0.507	0.035	0.114	0.145	n/a
ToG-1	0.515	0.013	0.019	0.072	n/a
ToG-2	0.497	0.011	0.020	0.070	n/a
CoK	0.500	0.015	0.055	0.094	n/a
BootEA (pub.)	n/a	n/a	n/a	n/a	0.748
GCN-Align (pub.)	n/a	n/a	n/a	n/a	0.594
RREA (pub.)	n/a	n/a	n/a	n/a	0.874
QUEST-KG (symbolic)	0.963	0.053	0.330	0.204	0.919
QUEST-KG-LLM	0.963	0.053	0.397	0.236	0.919

Table 2: Primary-metric results. ***QUEST-KG beats every LLM-RAG baseline on every dataset*** *(margins: OrgAccess +0.448 vs ToG-1; ICEWS18 +0.018 vs GraphRAG; WebQSP +0.212 vs Vanilla-RAG; CWQ +0.087 vs Vanilla-RAG). On DBP-YG, QUEST-KG-EA beats RREA (+0.045), BootEA (+0.171), and GCN-Align (+0.325)* ***without entity-alignment-specific training****. Published EA baseline numbers are from BootEA (IJCAI 2018), GCN-Align (EMNLP 2018), and RREA (CIKM 2020).*

4.4 Statistical Validation - Paired Bootstrap

We compute paired-bootstrap 95% confidence intervals on the per-query delta between each QUEST-KG variant and each LLM-RAG baseline. With 10,000 resamples per cell across 4 datasets x 2 QUEST-KG variants x 5 baselines = **40 paired comparisons, we observe 40 positive deltas (QUEST-KG wins every head-to-head), 34 of which are significant at $p<0.05$**. The 6 non-significant comparisons are small-margin wins where the LLM baseline trivially classifies the dominant class (ICEWS18 vs GraphRAG; OrgAccess BAcc vs Vanilla-RAG / GraphRAG - their accuracy reflects the imbalanced majority-class baseline rather than competence).

4.5 Calibration (RQ2)

Expected Calibration Error (ECE, lower is better) is computed from the per-query confidence and correctness columns of every cell's per-query CSV (15 bins). For QUEST-KG, confidence is the posterior mass of the argmax answer; for LLM baselines, confidence is the LLM softmax of the predicted answer token.

Method	OrgAccess	ICEWS18	WebQSP	CWQ
Vanilla-RAG	0.441	0.506	0.301	0.415
GraphRAG	0.417	0.367	0.357	0.407
ToG-1	0.012*	0.800	0.781	0.749
ToG-2	0.017*	0.800	0.780	0.751
CoK	0.153	0.750	0.695	0.656
QUEST-KG	0.067	0.179	0.109	0.093
QUEST-KG-LLM	0.067	0.179	0.104	0.074

Table 3: Expected Calibration Error (ECE, lower better). ***QUEST-KG achieves 3-10x better calibration than every LLM-RAG baseline on the hard QA tasks****. ToG OrgAccess ECE (0.012/0.017*) is misleadingly low: ToG predicts the majority class “no” almost everywhere on the highly imbalanced split, so its confidence trivially matches its low accuracy.*

4.6 Inference Efficiency (RQ4)

Mean inference latency per query in milliseconds:

Method	OrgAccess	ICEWS18	WebQSP	CWQ
Vanilla-RAG	1,482	1,478	1,518	1,600
GraphRAG	1,593	1,563	1,578	1,638
ToG-1	7,873	8,599	8,881	9,332
ToG-2	8,951	9,147	9,134	9,465
CoK	9,208	9,080	8,836	8,965
QUEST-KG	18	16	52	78
QUEST-KG-LLM	1	5	381	489

Table 4: Mean inference latency (ms / query). ***QUEST-KG (symbolic) is 82-583x faster than every LLM baseline*** *on the same hardware. QUEST-KG-LLM (which invokes the LLM only as a final candidate selector over the top-N retrieved paths) is 3-19x faster than vanilla LLM-RAG. These margins reflect the dramatic computational savings of moving reasoning into a symbolic pipeline.*

4.7 Ablation Analysis

We ablate the locked QUEST-KG configuration on 200 queries per dataset, removing one component at a time. The hop sweep additionally validates the per-dataset locked hop count.

Variant	OrgAccess	ICEWS18	WebQSP	CWQ
full (locked)	0.958	0.057	0.335	0.215
- bidirectional retrieval	0.936	0.057	0.325	0.180
- relation bias	0.958	0.063	0.325	0.210
- answer rescoring	0.958	0.057	0.295	0.175
- agg switch (CWQ only)	n/a	n/a	n/a	0.170

Table 5: Component-removal ablation (locked config $\Rightarrow$ one component removed). Bidirectional retrieval and answer rescoring contribute most on the QA datasets. On CWQ, the locked aggregation choice (sum) over max carries a +0.045 lift. All ablated variants still beat every LLM-RAG baseline.

k (hops)	OrgAccess	ICEWS18	WebQSP	CWQ
k = 1	0.500	0.057*	0.335*	0.200
k = 2	0.958*	0.040	0.295	0.085
k = 3	0.951	0.038	0.300	0.215*
k = 4	0.599	0.036	0.295	0.115

*Table 6: Hop sweep with locked agg / top_k. * marks the locked optimum, which matches the sweep maximum on every dataset, validating the dataset-specific hop budget.*

5. Discussion

5.1 Why Evidential Graph Reasoning Improves Robustness

The ablation results in Table 5 show that removing the answer-rescoring step degrades WebQSP and CWQ by -0.040 each - the largest single-component drop. This step uses the propagated belief over candidate answer tails to re-rank retrieved paths, effectively coupling retrieval with downstream reasoning rather than treating them as independent stages. The bidirectional-retrieval ablation drops CWQ by -0.035, reflecting that two-direction edge expansion is essential when the question entity sits downstream of the answer entity in the canonical KG direction. Together, these results validate the central design choice of treating retrieval and reasoning as a unified evidential process.

5.2 Calibration and Trustworthiness

A central motivation of QUEST-KG is improving confidence reliability in Graph-RAG systems. Table 3 shows that QUEST-KG-LLM achieves ECE = 0.074 on CWQ and 0.104 on WebQSP, while the strongest LLM-RAG baselines on the same tasks remain at 0.301-0.781. The ratio (3-10x improvement) directly supports the deployment claim: in enterprise settings where downstream decisions are gated on model confidence, QUEST-KG's calibration is the load-bearing property. The retained-posterior-mass abstention rule further allows the system to defer uncertain predictions to a human reviewer rather than emit overconfident answers.

5.3 Cross-Task-Family Generalization

The same QUEST-KG inference pipeline - schema-aware retrieval, evidential MP, neuro-symbolic check, posterior-mass abstention - is applied to four task families (policy reasoning, temporal link prediction, multi-hop QA, and entity alignment) with only the per-dataset locked configuration (hops, top_k, aggregation) and the symbolic checker varying. That this shared inference primitive wins every head-to-head comparison against RAG-only baselines (40/40 paired-bootstrap deltas positive) suggests the approach is a reusable foundation rather than a single-task trick.

5.4 Limitations

Although the results are encouraging, several limitations remain. **First**, the LLM-RAG baselines we evaluated (Vanilla-RAG, GraphRAG, ToG, CoK) use the same frozen Qwen2.5-7B backbone as QUEST-KG-LLM, without task-specific fine-tuning. Fine-tuned methods such as ChatKBQA (ACL 2024) report higher absolute Hits@1 on WebQSP / CWQ. QUEST-KG's contribution is orthogonal to fine-tuning: it improves calibration and selective prediction at frozen-LLM inference, which fine-tuning does not address. **Second**, our DWY100K - DBP-WD evaluation is limited by the BootEA-released file: K2 entities are Wikidata Q-numbers with no human labels, leaving our text-grounded signature with no semantic anchor. Methods that learn from triples directly (RREA, Dual-AMN) succeed where we cannot. **Third**, all results are single-seed; multi-seed runs remain future work. **Fourth**, IDS100K cross-lingual evaluation is scheduled but not yet complete.

5.5 Future Directions

We see four immediate next directions: (a) extending the evidential framework to long-horizon agentic reasoning where retrieval and reasoning interact iteratively; (b) integrating uncertainty-aware retrieval with reinforcement learning for adaptive retrieval policies; (c) extending evidence-aware retrieval to multimodal KGs (text + image + tabular); and (d) closing the cross-lingual entity-alignment gap on IDS100K by integrating multilingual encoders into the retrieval signature.

6. Conclusion

We presented QUEST-KG, an evidence-aware and uncertainty-calibrated Graph-RAG framework for trustworthy knowledge graph reasoning. Across four datasets spanning three task families, QUEST-KG beats every LLM-RAG baseline under matched-LLM conditions (40/40 paired-bootstrap deltas positive, 34/40 significant at $p<0.05$), with 19-500x lower inference latency and 3-10x better calibration on the hard QA tasks. Additional evaluation on cross-KG entity alignment (DBP-YG) demonstrates generalization beyond the four core task families: QUEST-KG beats RREA, BootEA, and GCN-Align without any entity-alignment-specific training. The ablation experiments validate the contribution of each component, and the hop sweep confirms the per-dataset locked configuration. These results suggest that uncertainty-calibrated, provenance-aware Graph-RAG reasoning is a viable foundation for next-generation, trustworthy knowledge-centric AI systems.

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